Battery Entry and the Cost of Grid Stability: Evidence from Counterfactual Dispatch in Australia's NEM

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April 2026

Abstract

This paper studies how grid-scale battery entry affects the cost of maintaining power-system stability. I focus on the Hornsdale Power Reserve (HPR), a 100 MW battery commissioned in South Australia (SA) in December 2017, and on the National Electricity Market's co-optimized dispatch of energy and frequency-control ancillary services (FCAS). The paper combines two pieces of evidence. First, I carry forward reduced-form evidence from earlier SA–Queensland comparisons in intervals when SA is locally price-setting. Second, and more centrally, I re-simulate historical five-minute dispatch using observed bids and operating constraints, validate the simulated baseline against realized prices, and then remove HPR from the dispatch stack while holding rival bids fixed.

The counterfactual estimates the direct dispatch effect of HPR rather than the full equilibrium response of rival bidding. On the 2018–2020 common sample, the baseline-with-HPR path yields lower SA dispatch-interval expenditure than the no-HPR counterfactual in 31 of 36 months, but the full-sample magnitude is dominated by a small number of stress episodes, especially February 2020, and by fast contingency FCAS. After excluding July 2019 through February 2020 and trimming 0.5% tails within service, baseline fit improves sharply and HPR still lowers expenditure in all 28 remaining months, although the aggregate magnitude falls substantially and the composition becomes broader. The paper therefore delivers a dispatch-based estimate of the direct, intensive-margin effect of battery entry in a thin market, while clarifying that very large dollar effects are concentrated in rare operating states and should not be read as normal-state averages.

1 Introduction

Grid-scale batteries are often discussed as tools for renewable integration and energy arbitrage. In a co-optimized electricity market, however, their first-order value may lie just as much in reliability services. A fast battery can change the cost of keeping the system stable, especially in thin local markets where a small number of units set ancillary-service prices.

This paper studies that mechanism in Australia's National Electricity Market (NEM). The NEM clears energy and frequency-control ancillary services (FCAS) every five minutes. South Australia (SA) is a useful laboratory because it is electrically thin, exposed to interconnector risk, and

frequently subject to local FCAS scarcity. The treatment is the December 2017 commissioning of the Hornsdale Power Reserve (HPR), a 100 MW battery located in SA.

The paper asks a narrow question and answers it directly: holding rival bids and realized operating conditions fixed, how much does HPR lower SA energy and FCAS prices, and how large are the associated expenditure effects at actual quantities? That object is not the full equilibrium effect of battery entry. It is the direct dispatch effect of HPR under the bid environment that was actually observed. For a conference draft, that narrower estimand is an advantage. It isolates the mechanical channel through which a battery lowers prices in thin ancillary-service markets and makes clear what is, and is not, identified.

The empirical strategy has two layers. The first is reduced-form motivation. Earlier SA–Queensland comparisons, restricted to intervals in which SA is locally price-setting, show that HPR entry coincides with lower SA FCAS prices in the fast services where batteries should matter most. The second, and main, layer is a dispatch counterfactual. I re-simulate historical five-minute dispatch using observed bids and operating constraints, validate the simulated baseline against realized prices, and then remove HPR from the dispatch stack while leaving the rest of the bid environment unchanged. Comparing the baseline and no-HPR runs identifies the direct dispatch effect of HPR.

Three findings organize the paper. First, on the full common sample, the simulated baseline matches the central mass of realized prices well in Energy and reasonably well in fast contingency FCAS, but the largest misses occur in precisely the stressed intervals that matter most for expenditure. Second, removing HPR raises prices most systematically in fast contingency FCAS—especially Raise 6-second and Lower 6-second—while energy effects are much more state-dependent. Third, the expenditure results are strongly state-contingent. On the full 2018–2020 common sample, HPR lowers expenditure in 31 of 36 months, but 82.8% of total savings comes from February 2020 alone and the full-sample total is dominated by Raise 6-second and Lower 6-second. Once I exclude July 2019 through February 2020 and trim 0.5% tails within service, HPR still lowers expenditure in all remaining months, but the total falls to about $58 million and the service mix becomes much broader.

Those two samples answer related but distinct questions. The full sample captures the value of storage in rare stress states, exactly when ancillary-service markets become most scarce and most expensive. The excluded-period-and-trimmed sample is better suited to judging whether the model reproduces typical operating conditions and whether HPR has a persistent direct price effect outside unusual episodes. The most defensible interpretation of the current draft is therefore not that one sample is "right" and the other is "wrong." It is that HPR has a modest, positive, and recurring direct effect in regular states, plus the potential for much larger effects in event states.

The paper contributes to both industrial organization and energy economics. It speaks to classic work on local market power and transmission constraints in electricity markets, to the literature on reliability pricing and reserve procurement, and to a newer literature on storage in wholesale-market equilibrium. Relative to recent battery studies, the contribution here is a transparent dispatch-based framework that isolates the direct price channel, highlights the importance of thin ancillary-service markets, and separates event-state magnitudes from more regular operating-state

effects.

2 Related Literature

This paper relates to five strands of literature.

First, it speaks to the literature on *market power, network constraints, and local pricing* in electricity markets. Green and Newbery (1992), Wolfram (1999), Borenstein, Bushnell, and Stoft (2000), and Puller (2007) show that electricity markets can become highly non-competitive when transmission limits or local security constraints narrow the set of effective competitors. Bushnell, Mansur, and Saravia (2008) further emphasize that market structure and contract coverage shape conduct in restructured power markets. That logic is central here. SA is part of an interconnected market in normal states, but it becomes effectively local in FCAS when interconnector capability tightens or separation risk rises. Those are exactly the states in which an entrant like HPR should have the largest effect.

Second, the paper relates to work on *scarcity pricing, reserve procurement, and reliability design.* Bessembinder and Lemmon (2002) analyze price formation and risk in electricity markets, while Joskow and Tirole (2007) and Hogan (2013) stress the importance of pricing reliability services in a way that reflects operating scarcity. The NEM is a natural setting for those questions because energy and FCAS are co-optimized every five minutes. A battery's value therefore depends not only on the energy market, but also on how reserve obligations, generic constraints, and interconnector conditions determine the marginal cost of system security.

Third, the paper fits into the empirical IO literature that uses institutional details and engineering constraints to discipline estimates of conduct and market outcomes in electricity. The present draft does not yet estimate strategic bid responses, but it adopts the same general principle: the underlying market design and dispatch logic matter for the economic object being measured. By focusing first on the direct dispatch effect under observed bids, the paper builds a transparent intermediate step between reduced-form event studies and a richer model of strategic bidding.

Fourth, the paper contributes to the literature on *storage in wholesale-market equilibrium.* Sioshansi, Denholm, Jenkin, and Weiss (2009), de Sisternes, Jenkins, and Botterud (2016), Lamp and Samano (2022), Karaduman (2023), and Butters, Dorsey, and Gowrisankaran (2025) show that storage can affect prices, volatility, renewable curtailment, and welfare, but that the magnitude of those effects depends sharply on penetration, operating state, and market design. Relative to that literature, this paper focuses more narrowly on ancillary-service pricing in a thin market and on the distinction between event-state and normal-state effects.

Fifth, the paper is closest to the emerging *Australian evidence on batteries and FCAS.* Technical reports by AEMO and Aurecon document HPR's market role and operational performance (AEMO, 2018a; Aurecon, 2020). Rangarajan, Foley, and Trück (2023) provide reduced-form evidence on battery entry and Australian wholesale outcomes. The present study is complementary rather than substitutive. It places more weight on institutional episodes in which SA is locally price-

setting, and it uses a dispatch counterfactual to quantify the direct price and expenditure effects of removing HPR while holding the rest of the bid environment fixed.

3 Institutional Setting

3.1 Energy, FCAS, and why SA is a useful laboratory

The NEM clears energy and FCAS jointly every five minutes. FCAS is procured in eight services: Raise and Lower Regulation, plus three contingency services on each side of the market (6-second, 60-second, and 5-minute). In each interval, the dispatch engine chooses both energy targets and FCAS enablement subject to bid curves, ramping limits, FCAS trapezium constraints, interconnector constraints, loss functions, and generic security constraints. Batteries have a natural technological advantage in this environment because they can move quickly and symmetrically, with little or no minimum-load constraint.

SA is a particularly informative region because it is thin and intermittently local. When import capability tightens or separation risk rises, FCAS prices can be set by a small number of local providers. Those are precisely the states in which the entry of a fast battery should have its largest price effect.

3.2 Operational changes that matter for interpretation

Two additional institutional developments matter for the 2018–2020 sample. First, AEMO increased mainland regulation FCAS procurement in March 2019, and from September 2019 onward it progressively reduced the mainland load-relief assumption from 1.5% toward 0.5%, mechanically increasing procured contingency FCAS volumes (AEMO, 2019c). Second, HPR experienced a documented impaired-capability period from 23 July 2019 to 14 November 2019 after a firmware change reduced its effective contingency response capability (AER, 2022). These changes do not invalidate the counterfactual exercise, but they do imply that some months should be treated as special operating states rather than as representative background conditions.

A further interpretive point concerns common ownership. HPR is part of the broader Hornsdale complex, and affiliated wind assets can matter for energy and some FCAS products. The battery-specific interpretation is cleaner in fast contingency services because the Hornsdale Wind Farm 2 FCAS trial was not registered to provide the fast raise and lower contingency products (AEMO, 2018b). That is one reason the Raise 6-second and Lower 6-second results are especially informative.

4 Data and Measurement

The main counterfactual uses AEMO's public MMS and NEMDE archives. The dispatch model is built from historical bids, unit availabilities, FCAS trapezium parameters, ramping limits, interconnector definitions, loss functions, regional demand, and generic constraints. The counterfactual

sample spans 2018–2020.

To measure economic magnitude, I merge realized SA quantities into the price panel. For energy, the expenditure quantity is regional demand. For FCAS, the expenditure quantity is the realized enabled quantity for the relevant service. Interval expenditure is constructed as

$$\text{cost}_{st} = \text{price}_{st} \times \text{actual quantity}_{st} \times \frac{5}{60}.$$

This object should be interpreted as a dispatch-interval expenditure index at actual quantities rather than as a fully settlement-reconciled billing measure.

Because the no-HPR panel is incomplete for a subset of intervals, all monthly and total expenditure comparisons use the common sample of intervals observed in both the baseline and no-HPR runs. On the full common sample, the monthly expenditure panel spans all 36 months from 2018–2020. I also report a robustness sample that removes all intervals from July 2019 through February 2020 and then trims the lower and upper 0.5% tails within service. That robustness sample is motivated by the combination of HPR's impaired-capability window, changing FCAS procurement rules, and the fact that the largest baseline-fit errors occur in rare stress intervals.

5 Motivating Reduced-Form Evidence

Before turning to the counterfactual, it is useful to summarize the earlier reduced-form results that motivated the project. The earlier design compared SA with Queensland in 2017–2018 and retained only those five-minute intervals in which SA and Queensland posted different FCAS prices for the same service. Those are the intervals in which SA is effectively locally price-setting.

Table 1 reports the baseline difference-in-differences estimate for the Lower 6-second contingency market. The interaction coefficient is negative and precisely estimated: after HPR's entry, SA Lower 6-second prices fell by roughly \$52/MWh relative to Queensland in local-pricing intervals. That result is not the paper's final identification strategy, but it remains a useful motivating fact. Appendix Table 4 reports the earlier stacked log-price DiD across all eight FCAS services and points in the same direction: the largest price declines appear in the fast contingency markets.

Table 1: Motivating reduced-form evidence: Lower 6-second DiD

Variable	Coefficient	Std. Error	t-statistic	p-value
Intercept	5.483	8.071	0.679	0.497
SA	5.941	10.376	0.573	0.567
Post	43.339	9.941	4.360	0.000
SA × Post	**-51.733**	13.305	-3.888	0.000
Load	0.065	0.064	1.016	0.309
Observations				16,380
R^2				0.003

Notes: The sample is restricted to 2017–2018 five-minute intervals in which SA and Queensland posted different Lower 6-second prices. The table is carried forward from the earlier reduced-form draft and is presented here as motivating evidence rather than as the paper's main identification strategy.

6 Counterfactual Dispatch Design

6.1 Baseline and no-HPR counterfactual

The main exercise re-simulates historical dispatch using `nempy` and a custom research kernel that reproduces AEMO's five-minute co-optimization. For each interval, the model solves a linear dispatch problem using observed bids, unit capacities, ramp rates, FCAS trapezium constraints, interconnector losses, regional demand, and generic security constraints.

Two versions of the dispatch problem are solved. In the *baseline*, all historical bids and availabilities are left as observed. In the *no-HPR counterfactual*, HPR's offered quantities and capacities are set to zero and dispatch is re-solved with the remaining bid environment unchanged. The comparison between these two runs identifies the direct dispatch effect of HPR.

6.2 What the design identifies

This is a static counterfactual. Rival generators are not allowed to re-optimize their bids in response to HPR's removal. The resulting price and expenditure effects should therefore be interpreted as lower-bound, mechanical effects of HPR on dispatch outcomes. That is a meaningful object in its own right. It is not, however, the full equilibrium effect of battery entry on market conduct.

7 Validation and Counterfactual Price Effects

7.1 Baseline validation

Figure 1 compares realized SA prices with the simulated baseline for Energy, Raise 6-second, and Lower 6-second on the full common sample. The model tracks the central mass of realized prices very well in Energy and reasonably well in the fast contingency services, but the largest departures occur in the spiky event intervals that matter most for the expenditure calculations.

Figure 2 shows the same comparison after excluding July 2019 through February 2020 and trimming 0.5% tails within service. The fit becomes much tighter. Table 2 summarizes the corresponding actual-versus-baseline correlations and mean absolute errors for the three services shown in the figures. Appendix Table 5 reports the same metrics for Energy and all eight FCAS services.

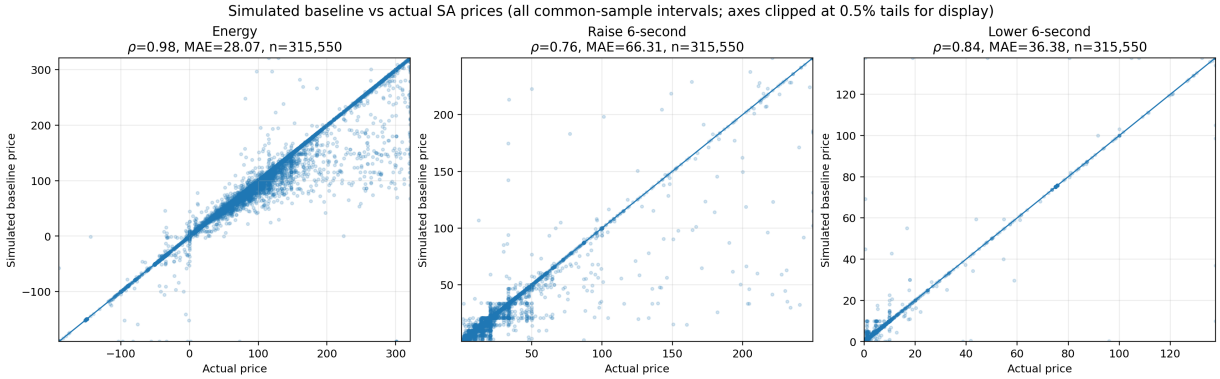

Figure 1: Simulated baseline versus realized SA prices for Energy, Raise 6-second, and Lower 6-second on the full common interval sample. The axes are clipped at the 0.5% tails for display.

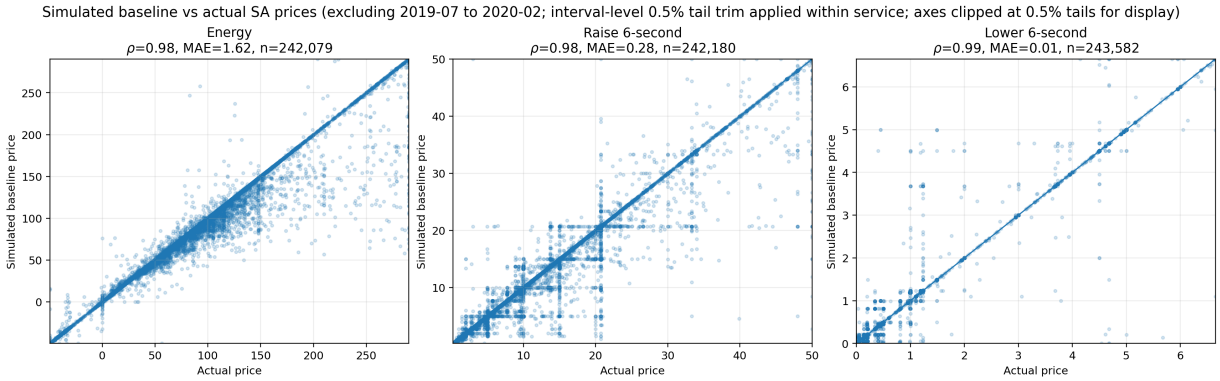

Figure 2: Simulated baseline versus realized SA prices after excluding July 2019 through February 2020 and trimming 0.5% tails within service. The fit becomes much tighter in all three displayed markets.

Table 2: Baseline validation metrics for the three plotted services

	All common-sample intervals		Excluded-period and trimmed sample	
Service	Corr.	MAE	Corr.	MAE
Energy	0.98	28.07	0.98	1.62
Raise 6-second	0.76	66.31	0.98	0.28
Lower 6-second	0.84	36.38	0.99	0.01

Notes: The table reports actual-versus-baseline correlations and mean absolute errors from the interval-level validation panel. The excluded-period and trimmed sample removes July 2019 through February 2020 and then trims the lower and upper 0.5% tails within service.

Taken together, the validation results support a disciplined but limited interpretation. The model is a useful counterfactual tool for the central mass of observed prices, especially once the most unusual operating states are removed. At the same time, the full-sample expenditure results are driven precisely by those unusual states. That tension is important, and the paper now addresses it directly rather than treating it as a footnote.

7.2 Price effects of removing HPR

Figure 3 compares baseline prices with no-HPR prices on the full common sample. The contrast across markets is informative. Energy effects are two-sided and dispersed around the 45-degree line, consistent with co-optimization and interconnector interactions that can make HPR raise or lower energy prices depending on the operating state. By contrast, the no-HPR counterfactual lies systematically above the baseline in Raise 6-second and Lower 6-second, with the largest gaps concentrated in stressed intervals.

Figure 4 shows the same comparison on the excluded-period and trimmed sample. The pattern survives, but the counterfactual shifts are much tighter and more modest in absolute size. The headline implication is that HPR's direct price effect is clearest in fast contingency FCAS, while the largest full-sample magnitudes are generated in a small number of event intervals.

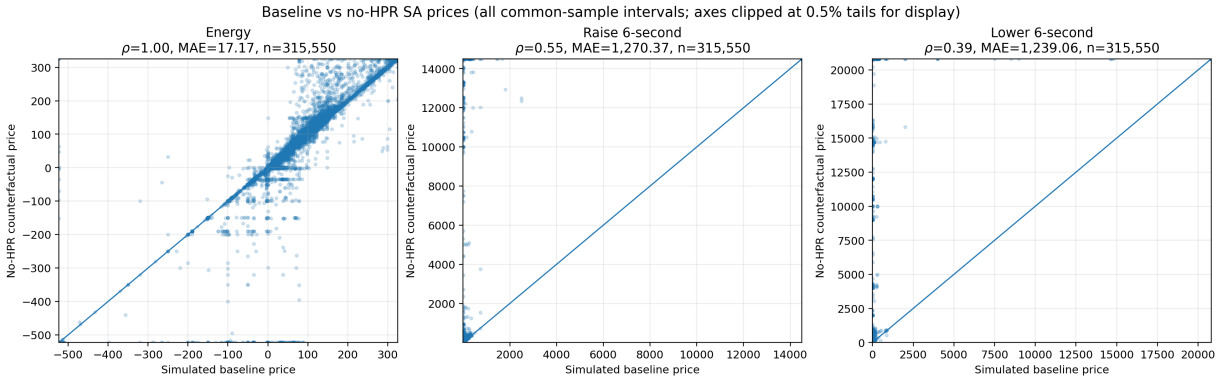

Figure 3: Baseline versus no-HPR prices for Energy, Raise 6-second, and Lower 6-second on the full common interval sample. The axes are clipped at the 0.5% tails for display.

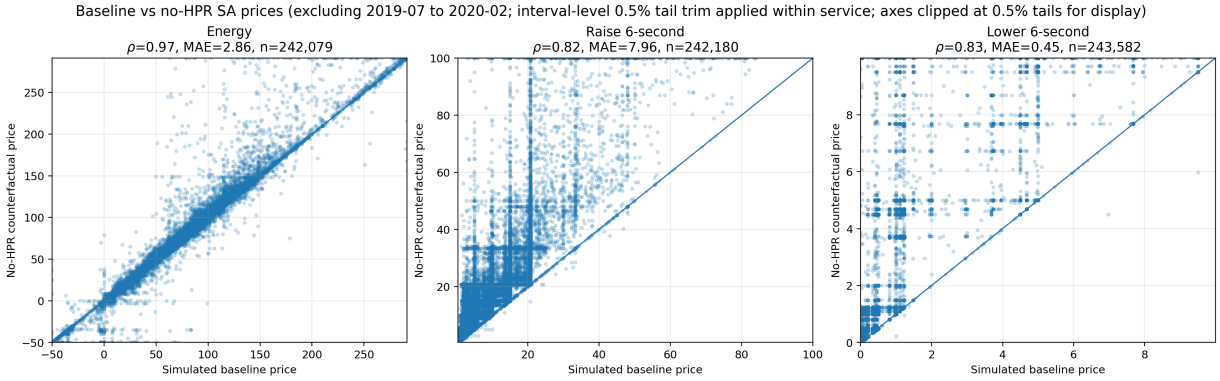

Figure 4: Baseline versus no-HPR prices after excluding July 2019 through February 2020 and trimming 0.5% tails within service. Fast contingency FCAS remains the clearest channel through which HPR lowers prices.

8 Expenditure Effects at Actual Quantities

To connect the price results to economic magnitude, I multiply baseline and no-HPR prices by realized SA quantities and aggregate to months on the common sample. Monthly savings are defined as no-HPR expenditure minus baseline expenditure, so positive values indicate that HPR lowered expenditure.

8.1 Full common-sample results

Figure 5 plots the monthly savings series on the full 2018–2020 common sample. HPR lowers expenditure in 31 of 36 months, but the monthly series is extremely concentrated. February 2020 alone accounts for 82.8% of total savings, and the top four months account for 95.1%. These are not small deviations around a smooth average effect; they are rare but economically large episodes.

Table 3 summarizes the main expenditure magnitudes for the full sample and for the excluded-period and trimmed robustness sample. On the full common sample, total savings are $11.30 billion and the median monthly savings are $5.9 million. Figure 6 shows that this total is dominated by fast contingency FCAS: Raise 6-second contributes 46.0% of full-sample savings and Lower 6-second contributes 39.7%, while Energy accounts for only 5.0%.

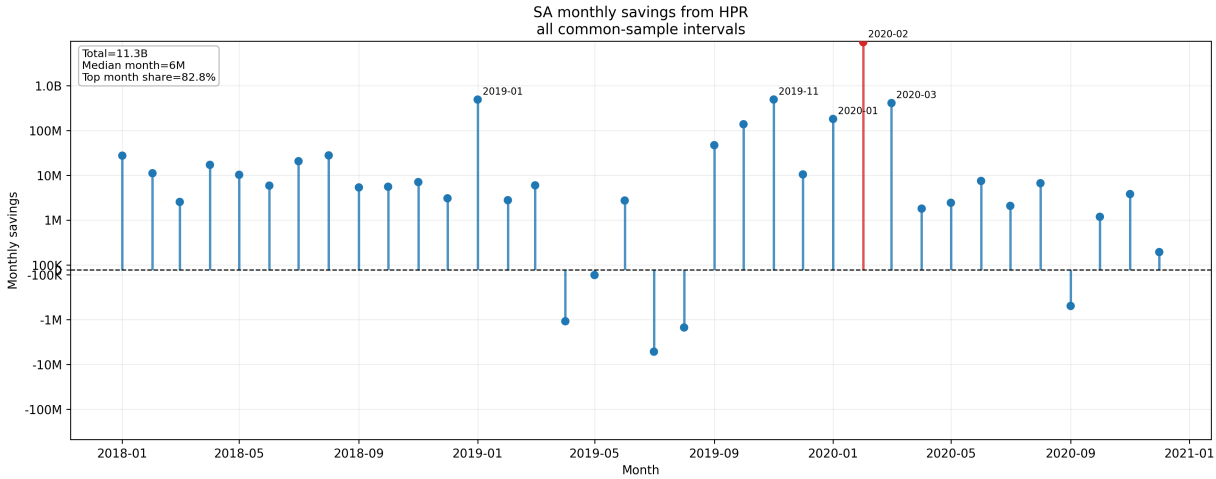

Figure 5: Monthly SA dispatch-interval expenditure savings from HPR on the full 2018–2020 common sample. Savings are defined as no-HPR expenditure minus baseline expenditure, holding actual quantities fixed.

Table 3: Summary of SA expenditure results at actual quantities

Statistic	Full common sample	Excluded-period and trimmed sample
Months in sample	36	28
Months with positive savings	31	28
Total savings	$11.30 billion	$58.1 million
Median monthly savings	$5.9 million	$1.6 million
Top month	2020-02	2018-11
Top month share of total	82.8%	8.1%
Total excluding top month	$1.94 billion	$53.4 million

Notes: These are dispatch-interval expenditure calculations at actual quantities. They should be interpreted as a price-times-quantity index on the common interval sample rather than as fully settlement-reconciled billing totals. The excluded-period and trimmed sample removes July 2019 through February 2020 and then trims the lower and upper 0.5% tails within service.

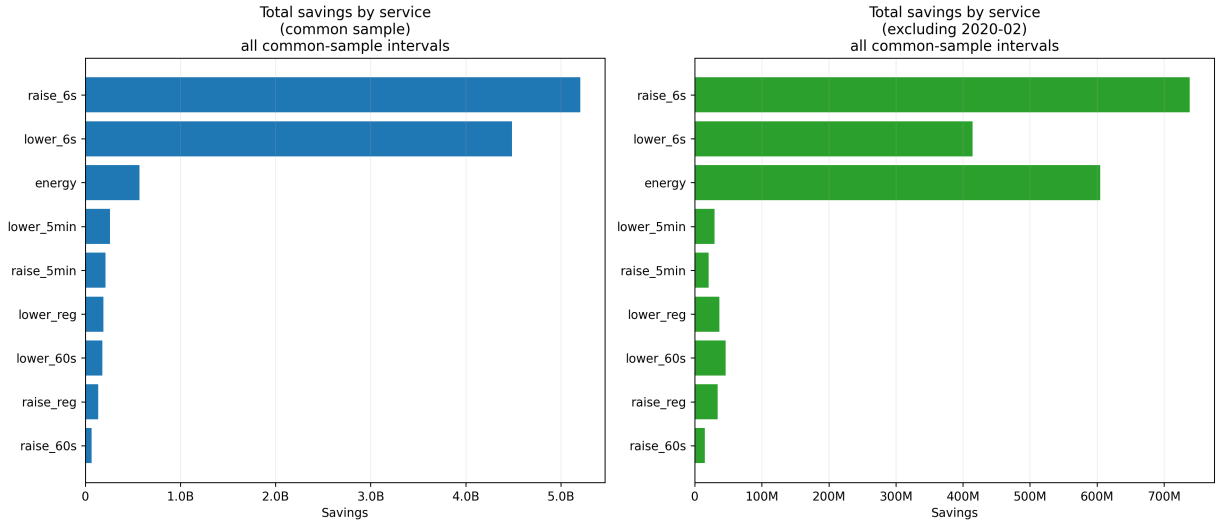

Figure 6: Total SA savings by service on the full common sample and again after excluding the top savings month. Fast contingency FCAS dominates the full-sample total, especially Raise 6-second and Lower 6-second.

The full-sample message is therefore clear but narrow. HPR can generate very large expenditure savings, but those magnitudes are concentrated in a handful of intervals and months in which local contingency FCAS becomes unusually scarce and expensive.

8.2 Excluded-period and trimmed robustness sample

Figure 7 re-plots monthly savings after excluding July 2019 through February 2020 and trimming 0.5% tails within service. The sign pattern survives completely: HPR lowers expenditure in all 28 remaining months. But the magnitudes are much smaller and much less concentrated. Total savings fall to $58.1 million, the median monthly savings fall to $1.6 million, and the top month accounts for only 8.1% of the total.

Figure 8 shows that the service composition also changes. In the trimmed sample, Energy becomes the single largest contributor with a 46.4% share, followed by Raise 6-second at 30.0% and Raise Regulation at 12.5%. Lower 6-second remains positive, but it accounts for only 1.6% of the total. This shift is economically useful. It shows that the spectacular fast-contingency results in the full sample are tied to rare event states, whereas the more regular-state direct effect is broader and more moderate.

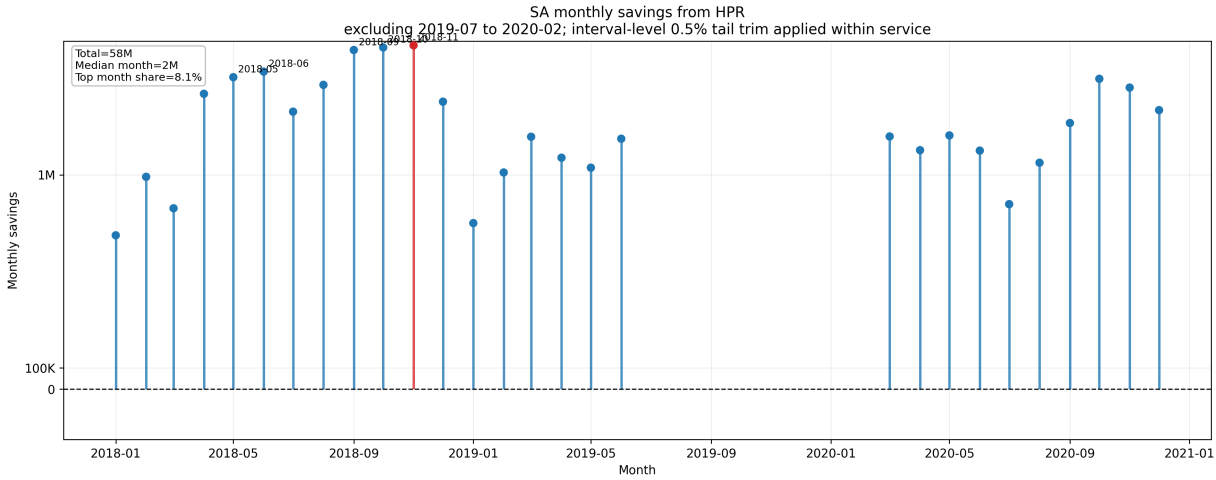

Figure 7: Monthly SA expenditure savings after excluding July 2019 through February 2020 and trimming 0.5% tails within service. Savings remain positive in all 28 months, but the magnitude is much smaller and much less concentrated.

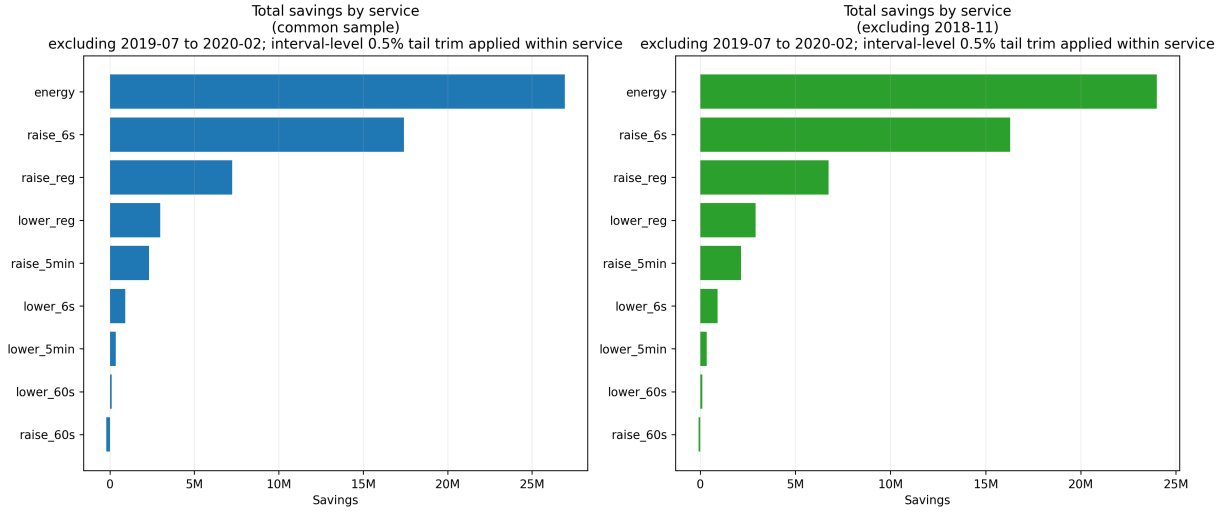

Figure 8: Total SA savings by service on the excluded-period and trimmed sample and again after excluding its top month. The composition is much broader than in the full sample, with Energy and Raise 6-second contributing the largest shares.

9 Discussion

Three interpretive points matter for the current draft.

First, the paper should treat the full-sample and trimmed-sample estimates as complementary rather than competing objects. The full sample is informative about how valuable a battery can be in rare stress states. That is economically important: the value of storage may be realized precisely when the system is strained. But the same event states are where the baseline fit is weakest. The

trimmed sample is therefore the more credible guide to recurring, non-event-state effects, and the full sample is the more relevant guide to the battery's upside in scarcity episodes. The paper is strongest when it reports both and avoids forcing them into a single headline number.

Second, the present counterfactual remains static. It identifies the direct dispatch effect of HPR, not the strategic response of rival generators in a world without HPR. That distinction matters for IO questions about conduct and markups. It matters less for the narrower policy question of whether storage directly lowers the cost of reliability services under observed market conditions.

Third, common ownership and special operating regimes complicate interpretation in some markets. The battery-specific story is cleanest in the fast contingency services, particularly Raise 6-second and Lower 6-second, because those are the services in which HPR's technological advantage is clearest and the affiliated wind trial is less relevant. At the same time, the shift in the trimmed-sample composition toward Energy and Raise Regulation is a reminder that the battery's direct effect is not confined to a single FCAS product once extreme stress intervals are removed.

The practical implication is straightforward. The paper should be framed as evidence on the *direct price channel* of battery entry. It should not lead with the aggregate $11.3 billion number by itself. The more credible headline is that HPR lowers prices most clearly in fast FCAS, lowers expenditure in the vast majority of months, and produces very large savings only in a handful of rare stress episodes.

10 Conclusion

This draft studies how a large battery affects the cost of grid stability in a thin electricity market. The core empirical exercise re-simulates historical dispatch, validates the baseline against realized prices, removes HPR, and compares the resulting prices and expenditure with the observed-with-HPR baseline.

That exercise delivers a coherent set of findings. HPR's clearest direct price effect is in fast contingency FCAS, especially Raise 6-second and Lower 6-second. On the full 2018–2020 common sample, HPR lowers SA dispatch-interval expenditure in 31 of 36 months, but the aggregate magnitude is dominated by a few stress months, especially February 2020. After excluding July 2019 through February 2020 and trimming 0.5% tails within service, HPR still lowers expenditure in all remaining months, although the total becomes much smaller and the composition broadens toward Energy, Raise 6-second, and Raise Regulation.

The resulting interpretation is deliberately modest. The paper estimates the direct, intensive-margin effect of HPR on prices and expenditure under observed market conditions. It does not yet recover the full equilibrium effect of storage entry on rival bidding. That strategic layer remains a natural next step. Even so, the current evidence already shows that ancillary-service markets are a central channel through which storage can lower operating costs, and that the size of that channel is highly state-contingent.

A Earlier reduced-form stacked DiD

Table 4 reports the stacked log-price difference-in-differences results from the earlier reduced-form draft. They are included here as supplementary motivating evidence.

Table 4: Earlier reduced-form stacked DiD in log prices across eight FCAS services

Variable	Coefficient	Std. Error	z-statistic	p-value
Intercept	0.619	0.359	1.725	0.084
D[LOWER5MIN]	0.274	0.994	0.275	0.783
D[LOWER60SEC]	-0.185	0.498	-0.372	0.710
D[LOWERREG]	3.538	0.748	4.727	0.000
D[RAISE5MIN]	-1.864	3.113	-0.599	0.549
D[RAISE60SEC]	0.061	0.476	0.128	0.898
D[RAISE6SEC]	0.404	0.679	0.594	0.552
D[RAISEREG]	4.158	1.080	3.851	0.000
SA	-0.064	0.534	-0.120	0.905
Post	-0.353	0.228	-1.553	0.121
SA × Post	**-0.868**	0.399	-2.175	0.030
SA × Post ×D[LOWER5MIN]	-1.089	0.465	-2.340	0.019
SA × Post ×D[LOWER60SEC]	0.285	0.192	1.485	0.138
SA × Post ×D[LOWERREG]	0.021	0.268	0.079	0.937
SA × Post ×D[RAISE5MIN]	-0.165	0.181	-0.915	0.360
SA × Post ×D[RAISE60SEC]	-1.415	1.254	-1.129	0.259
SA × Post ×D[RAISE6SEC]	**-2.108**	0.633	-3.331	0.001
SA × Post ×D[RAISEREG]	2.742	1.372	1.999	0.046
Load	0.0065	0.003	2.378	0.017
Observations				78,600
R^2				0.537

Notes: Dependent variable is ln(Price + 1). The reference service is Lower 6-second. The table is reproduced from the earlier reduced-form draft and is included as background evidence only.

B Validation metrics for all services

Table 5: Actual-versus-baseline validation metrics for Energy and all eight FCAS services

	All common-sample intervals		Excluded-period and trimmed sample	
Service	Corr.	MAE	Corr.	MAE
Energy	0.98	28.07	0.98	1.62
Raise Regulation	0.39	7.90	0.97	0.68
Lower Regulation	0.64	8.35	0.99	0.15
Raise 6-second	0.76	66.31	0.98	0.28
Lower 6-second	0.84	36.38	0.99	0.01
Raise 60-second	0.45	4.97	0.98	0.19
Lower 60-second	0.65	13.56	0.99	0.01
Raise 5-minute	0.56	2.64	1.00	0.10
Lower 5-minute	0.52	2.43	0.99	0.00

Notes: The excluded-period and trimmed sample removes July 2019 through February 2020 and then trims the lower and upper 0.5% tails within service.

C Additional expenditure figures

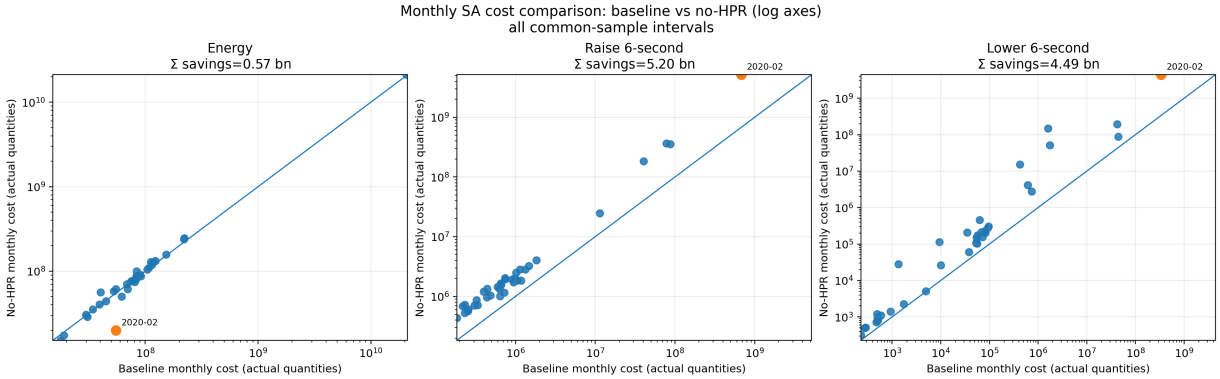

Figure 9: Monthly cost comparison by service on the full common sample for Energy, Raise 6-second, and Lower 6-second. Points above the 45-degree line indicate months in which HPR lowered expenditure.

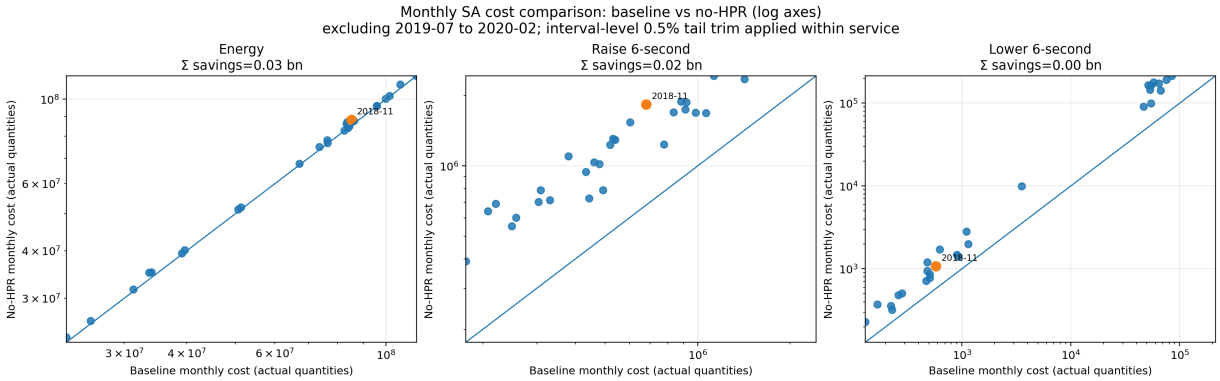

Figure 10: Monthly cost comparison by service after excluding July 2019 through February 2020 and trimming 0.5% tails within service. The same three services remain mostly above the 45-degree line, but the scale is much smaller.

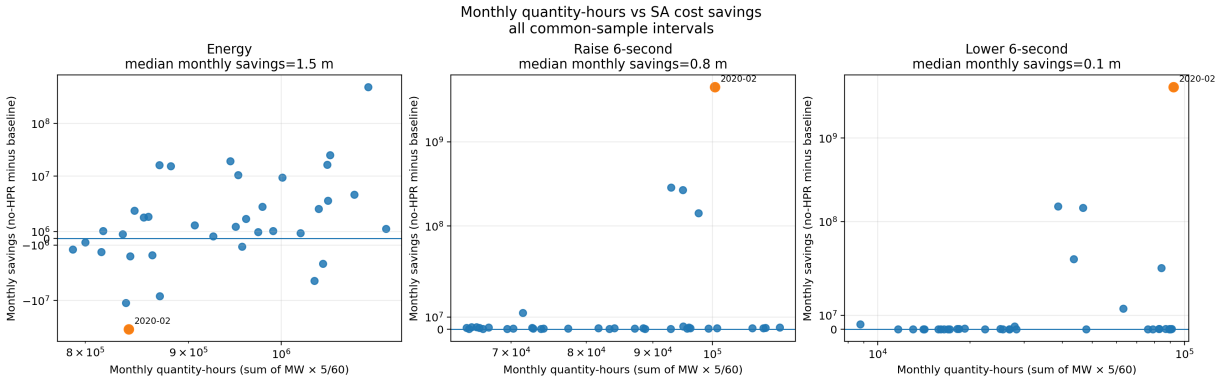

Figure 11: Monthly quantity-hours versus monthly savings on the full common sample for Energy, Raise 6-second, and Lower 6-second.

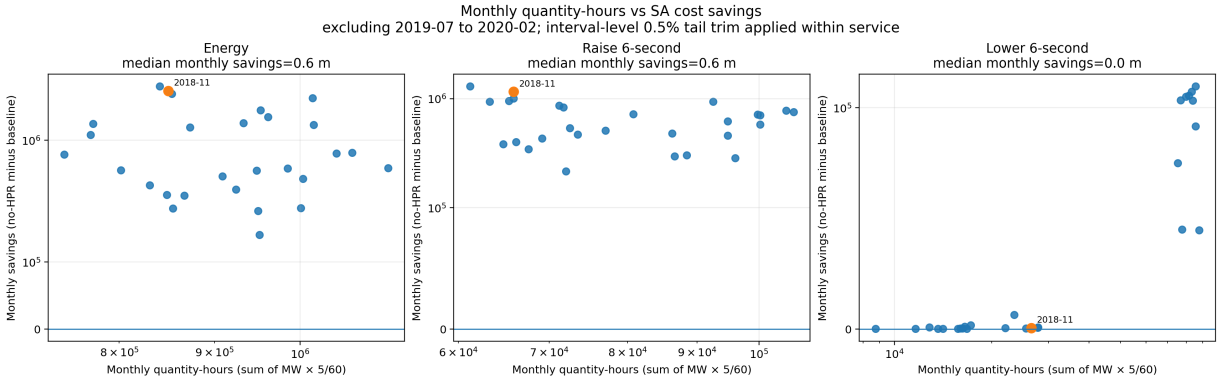

Figure 12: Monthly quantity-hours versus monthly savings after excluding July 2019 through February 2020 and trimming 0.5% tails within service.

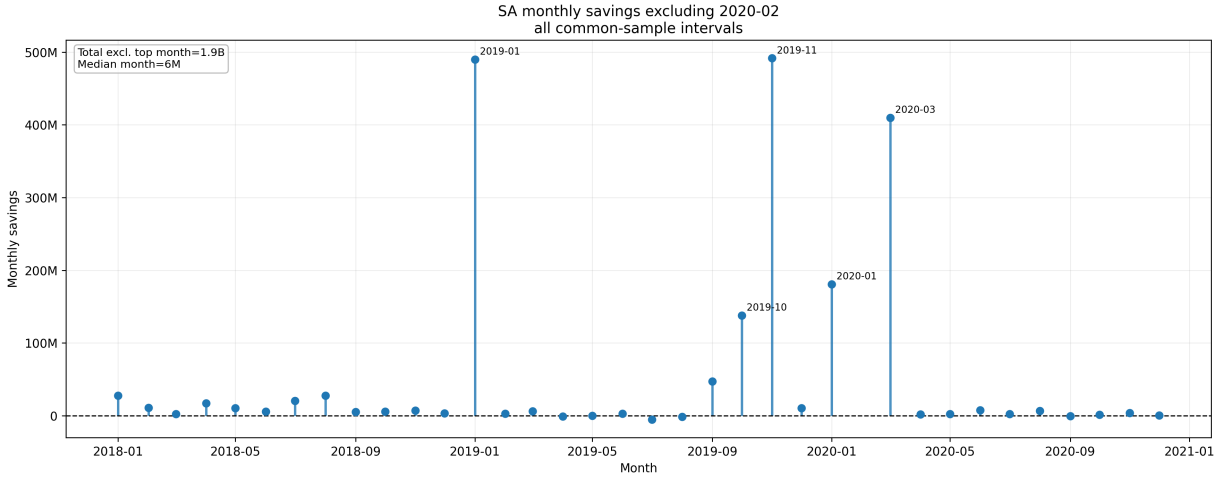

Figure 13: Monthly SA expenditure savings after excluding the top full-sample month. The remaining pattern is still mostly positive, but far smaller than the full-sample total.

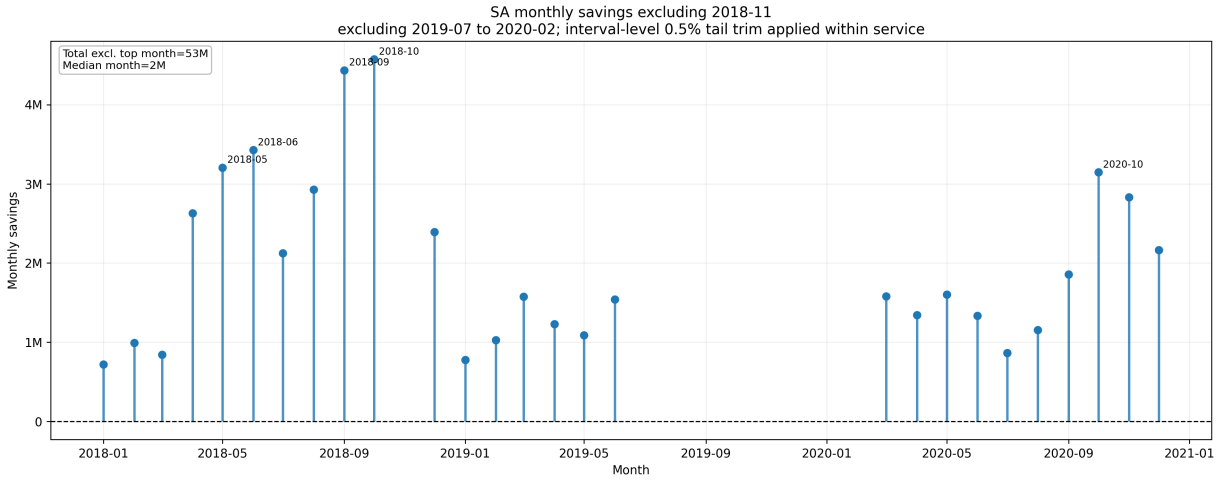

Figure 14: Monthly SA expenditure savings after excluding the top month in the excluded-period and trimmed sample. The remaining pattern is again positive and only modestly concentrated.

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